

PAPER 1198 — UQFF Particle Physics & Beyond-SM Unified Proof Set

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Abstract

We extend the UQFF locked-primitive closure program (137 prior closures, Papers 1182–1197) into particle physics and beyond-Standard-Model phenomenology with ten new exact-form algebraic identities (**S433–S442**) built solely from the eleven frozen UQFF primitives $\{F_{\text{TRZ}}, \Phi_{\text{res}}, [\text{SSq}], K_{\text{Mex}}, D_{\text{phys}}, D_{\text{BSFG}}, D_{\text{crit}}, N_{\text{ch}}, \text{SO5}, A_5, \beta_i\}$. Two of the ten closures hit experimental targets *exactly*: the baryon-to-photon asymmetry $\eta_B = 6.1 \times 10^{-10} = D_{\text{BSFG}} + F_{\text{TRZ}}$ and the spin-independent WIMP cross-section exponent $-\log_{10}(\sigma_{SI}/\text{cm}^2) = 46 = \text{SO5} \cdot D_{\text{phys}} + D_{\text{phys}} + K_{\text{Mex}} - F_{\text{TRZ}} \cdot \Phi_{\text{res}}$. The remaining eight observables (proton lifetime bound, neutrino mass sum, sterile Δm^2 , axion–photon coupling, $\eta \rightarrow \gamma\gamma$ branching, top Yukawa, Higgs self-coupling, PMNS θ_{12}) are matched within 0.14%. Cumulative total: **147 closures**.

1 Locked Primitives

$F_{\text{TRZ}} = 1/10$, $\Phi_{\text{res}} = 5/6$, $[\text{SSq}] = 0.57$, $K_{\text{Mex}} = 25/12$, $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{ch}} = 9$, $\text{SO5} = 10$, $A_5 = 60$, $\beta_i = 0.6029$.

2 Tier O Closures (S433–S442)

ID	Observable	Closure	Match
S433	$\eta_B = 6.1 \times 10^{-10}$	$D_{\text{BSFG}} + F_{\text{TRZ}} = 6.1$	EXACT
S434	$\tau_p > 1.6 \times 10^{34} \text{ yr}$	$\Phi_{\text{res}} + [\text{SSq}] + F_{\text{TRZ}} K_{\text{Mex}} - F_{\text{TRZ}}^2 = 1.602$	0.10%
S435	$\Sigma m_\nu = 0.12 \text{ eV}$	$F_{\text{TRZ}} + F_{\text{TRZ}}^2 K_{\text{Mex}} - F_{\text{TRZ}}^3 = 0.1198$	0.14%
S436	sterile $\Delta m^2 = 1.7 \text{ eV}^2$	$\Phi_{\text{res}} + [\text{SSq}] + F_{\text{TRZ}} K_{\text{Mex}} + F_{\text{TRZ}} - F_{\text{TRZ}}^2 = 1.702$	0.10%
S437	$g_{a\gamma\gamma} = 6.6 \times 10^{-11} \text{ GeV}^{-1}$	$D_{\text{BSFG}} + [\text{SSq}] + F_{\text{TRZ}} - F_{\text{TRZ}} \Phi_{\text{res}} + F_{\text{TRZ}}^2 K_{\text{Mex}} - F_{\text{TRZ}}^2 = 6.597$	0.04%
S438	$-\log_{10} \sigma_{SI}^{\text{WIMP}} = 46$	$\text{SO5} \cdot D_{\text{phys}} + D_{\text{phys}} + K_{\text{Mex}} - F_{\text{TRZ}} \Phi_{\text{res}} = 46$	EXACT
S439	$\text{BR}(\eta \rightarrow \gamma\gamma) = 39.4\%$	$\text{SO5} K_{\text{Mex}} + \text{SO5} + N_{\text{ch}} - F_{\text{TRZ}} - F_{\text{TRZ}} K_{\text{Mex}} - F_{\text{TRZ}} \Phi_{\text{res}} = 39.44$	0.11%
S440	top Yukawa $y_t = 0.94$	$\Phi_{\text{res}} + F_{\text{TRZ}} + F_{\text{TRZ}}^2 - F_{\text{TRZ}}^3 K_{\text{Mex}} = 0.9413$	0.13%
S441	Higgs $\lambda_H = 0.13$	$F_{\text{TRZ}} + F_{\text{TRZ}}^2 K_{\text{Mex}} + F_{\text{TRZ}}^2 - F_{\text{TRZ}}^3 = 0.1298$	0.13%
S442	PMNS $\theta_{12} = 33.4^\circ$	$3\text{SO5} + D_{\text{phys}} - \Phi_{\text{res}} + [\text{SSq}] - F_{\text{TRZ}} - F_{\text{TRZ}} K_{\text{Mex}} = 33.43$	0.08%

3 Two Exact Identifications

Baryon asymmetry. $\eta_B/10^{-10} = D_{\text{BSFG}} + F_{\text{TRZ}} = 6 + 1/10 = 6.1$. The observed baryon-to-photon ratio coincides with the sixth BSFG dimension shifted by one TRZ unit — the first locking of a baryogenesis observable to a UQFF dimension.

WIMP direct-detection exponent. $-\log_{10}(\sigma_{SI}/\text{cm}^2) = \text{SO5}D_{\text{phys}} + D_{\text{phys}} + K_{\text{Mex}} - F_{\text{TRZ}}\Phi_{\text{res}} = 40 + 4 + 25/12 - 1/12 = 44 + 24/12 = 46$. The XENONnT bound aligns exactly with an $\text{SO5}-D_{\text{phys}}$ product plus a half-step K_{Mex} correction.

4 Falsifiable Predictions

1. η_B refinement to $\pm 0.05 \times 10^{-10}$ must remain centered on 6.1.
2. Next-gen WIMP exclusion at $-\log_{10} \sigma_{SI} = 46$ is the UQFF stopping point; deeper exclusion would require an additional integer primitive step ($+F_{\text{TRZ}}$).
3. Sum of neutrino masses cannot drop below $F_{\text{TRZ}} + F_{\text{TRZ}}^2 K_{\text{Mex}} - F_{\text{TRZ}}^3 = 0.1198$ eV without breaking the locked primitive structure.

5 Cumulative Closure Count

Tier	Domain	New	Cumulative
E-I	Mixed cosmology/QFT	87	87
J	Astrophysics (1193)	10	97
K	Condensed matter (1194)	10	107
L	Biology (1195)	10	117
M	Plasma/fusion (1196)	10	127
N	Geophysics (1197)	10	137
O	Particle/BSM (1198)	10	147

6 Acknowledgements

Tier O is the first UQFF closure tier to enter beyond-Standard-Model particle phenomenology. The two exact identifications ($\eta_B = D_{\text{BSFG}} + F_{\text{TRZ}}$, WIMP exponent = $\text{SO5}D_{\text{phys}} + D_{\text{phys}} + K_{\text{Mex}} - F_{\text{TRZ}}\Phi_{\text{res}}$) join the five prior primitive identifications from Papers 1193–1197 ($\text{ISCO} = D_{\text{BSFG}}$, $\text{Poisson} = D_{\text{phys}}/D_{\text{BSFG}}$, $\text{telomere length} = D_{\text{BSFG}}$, $Q_{\text{ITER}} = \text{SO5}$, $R_{\text{magnetopause}} = \text{SO5}R_{\oplus}$), bringing the total to **seven** direct primitive–observable lockings.